

Taehyeon Kim

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RESEARCH PROFILE

- **Efficient reasoning and agent learning:** Research on adaptive thinking, reinforcement learning, inference-time control, and fast decoding.
- **Embodied AI:** Senior Member of Technical Staff at Holiday Robotics, developing agent harnesses, RL action policies, data flywheels, and evaluation.
- **Research leadership:** Former Tech Lead and Squad Lead at LG AI Research; led 10+ researchers (6+ PhDs) on agentic search, tool use, and LLM alignment.

SELECTED RESEARCH

Root-token Policy Optimization (RPO) [C15] – first author, **ACL 2026 (Oral)**. Paper

- Proposed an RL framework that uses group-relative rewards to optimize the initial root-token decision: whether to produce an explicit reasoning trace.
- Learned difficulty-aware adaptive thinking at approximately **2% of the training compute** of the prior adaptive-reasoning methods evaluated, across multiple model families and scales.

Instructive Decoding [C7] – co-first author, **ICLR 2024 Spotlight**. Paper

- Proposed a training-free method that contrasts next-token predictions with predictions from noisy instructions to improve instruction following, without updating model parameters.

Blockwise parallel decoding with draft refinement [C10] – first author, **NeurIPS 2024**. Paper

- Analyzed predictions across multiple token heads and proposed draft-refinement algorithms using n-gram and neural language-model rescoring.
- Increased mean accepted token length by **5–25% relative**, yielding **over 3x wall-clock speedup** over standard autoregressive decoding on the evaluated open-source 7B and 13B LLMs.

Additional selected work

- **Contrastive Thinking Decoding [C16]** – first author, **NeurIPS 2026**. Steering answer generation in reasoning models.
- **Multi-Drafter Speculative Decoding [C14]** – co-first author, **Findings of ACL 2026**. Multiple drafters with alignment feedback for speculative decoding.
- **Knowledge distillation [C2]** – co-first author, **IJCAI 2021**. Comparing Kullback-Leibler divergence and mean squared error objectives for knowledge distillation.

WORK EXPERIENCE

Holiday Robotics

Jun 2026 - Present

Senior Member of Technical Staff, Embodied AI Team

- **Harness development and learning:** Developing planning, perception for spatial intelligence, and act-verify loops in agent execution harnesses, alongside meta-harness-style training with dense feedback. Founding member of the Embodied AI Team.
- **Text-and-motion-conditioned RL policies:** Developed an algorithm that automates manual engineering in constructing RL action policies from text and motion. Behavior-cloning regularization helps the resulting policies follow intended behaviors.
- **Data flywheel:** Working on grasping and video-based progress labeling across real-world and simulated robot data.
- **Evaluation (in progress):** Building benchmarks for embodied-agent behavior and task execution.

LG AI Research

Apr 2025 - Jun 2026

Superintelligence Lab: Research Scientist (Apr-Aug 2025); Squad Lead (Sep-Dec 2025); Tech Lead (Jan-Jun 2026)

- **Research leadership:** Led a team of **10+ researchers (6+ PhDs)** on agentic search and LLM alignment; defined the research roadmap, mentored researchers, and coordinated project delivery.
- **Agentic search:** Built an end-to-end search system over internal expert documents, from retrieval to agentic reasoning. Achieved higher accuracy than Perplexity, Codex, and Anthropic systems in internal evaluations on this task.
- **Agent systems and multimodal learning:** Built multi-tool orchestration with parallel retrieval, query planning, and examination tools. Trained dense retrievers and rerankers with hard-negative mining; designed VLM experiments for diagram understanding and retrieval grounding.
- **LLM alignment and RL:** SFT, DPO, and GRPO-based training; first-author RPO (ACL 2026, Oral [C15]); small-large model collaboration; Masked-SIMPER (in preparation [U3]).
- **Research:** Contrastive Thinking Decoding (NeurIPS 2026 [C16]); co-corresponding author on AdaSTaR (NeurIPS 2025 [C12]); MERIT negotiation benchmark (ACL 2026 [C13]); multi-drafter speculative decoding (Findings of ACL 2026 [C14]).

Google Research

Oct 2023 - Dec 2023

PhD Research Intern; hosts: Adrian Benton, Ananda Theertha Suresh, and Kishore Papineni

- Developed fast parallel decoding methods for LLMs; first-author NeurIPS 2024 paper [C10].

Dynamo AI (YC-backed)

Jan 2023 - May 2023

PhD Research Intern; worked with Eric Lin

- Privacy-preserving federated learning for enterprise AI; contributed to automotive B2B contracts and a first-author NeurIPS 2023 paper [C5].

National Institute of Meteorological Sciences (NIMS)

Mar 2022 - Jun 2023

Research Manager

- Transformer-based precipitation forecasting; NeurIPS 2022 Weather4Cast competition award, collaborating with a team of 40+ researchers.

Qualcomm AI

Jun 2021 - Dec 2021

PhD Research Intern; worked with Heesoo Myeong

- Resource-efficient backbone design for ADAS; ICML 2022 HAET Workshop [W2] and U.S. patent [P1].

EDUCATION**Ph.D., Graduate School of AI, KAIST**

Mar 2020 - Feb 2025

Advisor: Prof. Se-Young Yun

Thesis: Efficient and Effective Inference-time Decoding Strategies for Foundational Models.

M.S., Graduate School of Data Science, KAIST

Mar 2018 - Feb 2020

Advisor: Prof. Se-Young Yun

Thesis: Orthogonal Feature Regularization: A Novel Approach for Training Robust Models.

B.S., Mathematical Sciences, KAIST

Mar 2013 - Feb 2018

Minor: Intellectual Property Program

PUBLICATIONS

* equal contribution; † corresponding authors.

- C16 **Kim, T.**, Jang, Y., Lee, H., Kim, Y. J. & Lee, M. (2026). Contrastive Thinking Decoding: Steering Answer Generation in Reasoning Models, **NeurIPS 2026**.
- C15 **Kim, T.**, Lee, H., Jang, Y. & Lee, M. (2026). Efficiently Learning To Reason or Not to Reason: Root-token Policy Optimization for Adaptive Thinking, The 64th Annual Meeting of the Association for Computational Linguistics (**ACL 2026**) (**Oral**), Jul. 2026.
- C14 **Kim, T.***, Jung, H.* & Yun, S. (2026). Multi-Drafter Speculative Decoding with Alignment Feedback, The 64th Annual Meeting of the Association for Computational Linguistics (**ACL 2026 Findings**), Jul. 2026.
- C13 Oh, J., Aghazada, M., Shin, Y., Yun, S. & **Kim, T.†** (2026). MERIT Feedback Elicits Better Bargaining in LLM Negotiators, The 64th Annual Meeting of the Association for Computational Linguistics (**ACL 2026**), Jul. 2026.
- W9 Kim, Y., Yi, E., Kim, M., Yun, S.†, **Kim, T.†** (2025). Guiding Reasoning in Small Language Models with LLM Assistance, **CoLM 2025 Workshop: Test-time Scaling and Reasoning Models**.
- C12 Koh, W., Oh, W., Jang, J., Lee, M., Kim, H., Kim, A., Kim, J., Lee, J., **Kim, T.†**, Yun, S.† (2025). AdaSTaR: Adaptive Data Sampling for Training Self-Taught Reasoners, **Advances in Neural Information Processing Systems 38 (NeurIPS 2025)**, Dec. 2025.
- C11 Koh, W., Yoon, J., Lee, M., Song, Y., Cho, J., Kang, J., **Kim, T.**, Yun, S., Yu, Y., Lee, B. (2025). C²: Scalable Auto-Feedback for LLM-based Chart Generation, **The 2025 Annual Conference of the Nations of the Americas Chapter of the ACL (NAACL 2025) (Oral)**, May. 2025.
- J2 **Kim, T.**, Kim, D. & Yun, S. (2024). FLR: Label-Mixture Regularization for Federated Learning with Noisy Labels, Transactions on Machine Learning Research (**TMLR**).
- W8 **Kim, T.***, Jung, H.* & Yun, S. (2024). A Unified Framework for Speculative Decoding with Multiple Drafters as a Bandit, NeurIPS 2024 Workshop: Efficient Natural Language and Speech Processing (ENLSP-IV).
- C10 **Kim, T.**, Suresh, AT., Papineni, K., Riley, M., Kumar, S. & Benton, A. (2024). Accelerating Blockwise Parallel Language Models with Draft Refinement, **Advances in Neural Information Processing Systems 37 (NeurIPS 2024)**, Dec. 2024.
- C9 Ho, N.*, Bae, S.*, **Kim, T.**, Jo, H., Kim, Y., Schuster, T., Fisch, A., Thorne, J. and Yun, S. (2024). Block Transformer: Global-to-Local Language Modeling for Fast Inference, **Advances in Neural Information Processing Systems 37 (NeurIPS 2024)**, Dec. 2024.
- C8 Yi, E.*, **Kim, T.***, Jeung, H., Chang, D. and Yun, S. (2024). Towards Fast Multilingual LLM Inference: Speculative Decoding and Specialized Drafters, **The 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP 2024)**, Nov. 2024.
- W7 **Kim, T.**, Suresh, AT., Papineni, K., Riley, M., Kumar, S. & Benton, A. (2024). Exploring and Improving Drafts in Blockwise Parallel Decoding, ICML 2024 Workshop: Efficient Systems for Foundation Models (Es-FoMo).
- C7 **Kim, T.***, Kim, J.*, Lee, G.* & Yun, S. (2024). Instructive Decoding: Instruction-Tuned Large Language Models are Self-Refiner from Noisy Instructions, **Twelfth International Conference on Learning Representations (ICLR 2024) (Spotlight, Top 5%)**, May. 2024.
- C6 Yang, Y., **Kim, T.** & Yun, S. (2024). Leveraging Normalization Layer in Adapters with Progressive Learning and Adaptive Distillation for Cross-Domain Few-Shot Learning, **The 38th Annual AAAI Conference on Artificial Intelligence (AAAI 2024)**, Feb. 2024.

- C5 **Kim, T.**, Lin, E., Lee, J., Lau, C. & Mugunthan, V. (2023). Navigating Data Heterogeneity in Federated Learning: A Semi-Supervised Federated Object Detection, *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, Dec. 2023.
- C4 Gruca et al. Weather4cast at NeurIPS 2022: Super-Resolution Rain Movie Prediction under Spatio-temporal Shifts, *Proceedings of the NeurIPS 2022 Competition and Demonstration Track, in Proceedings of Machine Learning Research 220:292-313 (2022)*.
- W6 **Kim, T.**, Kang, S., Shin, H., Yoon, D., Eom, S., Shin, K. & Yun, S. (2022). Region-Conditioned Orthogonal 3D U-Net for Weather4Cast Competition, NeurIPS 2022 Workshop: Weather4Cast Competition.
- W5 Eom, S., **Kim, T.** & Yun, S. (2022). Layover Intermediate Layer for Multi-Label Classification in Efficient Transfer Learning, NeurIPS 2022 Workshop: Has it Trained Yet? (HITY).
- W4 Shin, J., **Kim, T.** & Yun, S. (2022). Revisiting the Activation Function for Federated Image Classification, NeurIPS 2022 Workshop Federated Learning.
- W3 **Kim, T.** & Yun, S. (2022). Supernet Training for Federated Image Classification under System Heterogeneity, ICML 2022 Workshop on Dynamic Neural Networks (**Oral**).
- J1 **Kim, T.** & Yun, S. (2022). Revisiting Orthogonality Regularization: A Study for Convolutional Neural Networks in Image Classification, *IEEE Access*, Jun. 2022.
- W2 **Kim, T.**, Myeong, H. & Yun, S. (2022). Revisiting Architecture-aware Knowledge Distillation: Smaller Models and Faster Search, ICML 2022 Hardware Aware Efficient Training (HAET) Workshop, July 2022.
- C3 **Kim, T.***, Ko, J.*, Cho, S., Choi, J. & Yun, S. (2021). FINE Samples for Learning with Noisy Labels, *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, Dec. 2021.
- C2 **Kim, T.***, Oh, J.*, Kim, N., Cho, S., & Yun, S. Y. (2021). Comparing Kullback-Leibler Divergence and Mean Squared Error Loss in Knowledge Distillation. *In the 30th International Joint Conference on Artificial Intelligence (IJCAI 2021)*, Aug. 2021 (acceptance rate: 13.9%)
- W1 **Kim, T.***, Ahn, J.*, Kim, N.*, & Yun, S. (2020). Adaptive Local Bayesian Optimization Over Multiple Discrete Variables. Workshop at NeurIPS 2020 Competition Track on Black-Box Optimization Challenge, Dec. 2020.
- C1 **Kim, T.**, Kim, J. & Yun, S. (2019). Efficient Model for Image Classification With Regularization Tricks. *Proceedings of the NeurIPS 2019 Competition and Demonstration Track, in Proceedings of Machine Learning Research 123:13-26*.
- T **Kim, T.**. Orthogonal feature regularization : a novel approach for training robust models, Korea Advanced Institute of Science and Technology (KAIST)

ONGOING MANUSCRIPTS

- U3 **Kim, T.** et al. (2025). Masked-SIMPER: Improving Collaborative Reasoning, In preparation.
- U1 Bartholet, M.*, **Kim, T.***, Beuret, A. Yun, S. & Buhmann, J. (2024). Hypernetwork-Driven Model Fusion for Federated Domain Generalization, Under review.

US PATENTS

- P1 **Kim, T.**, Myeong, H. (2023). Trust-Region Aware Architecture Distillation for Sample-Efficient Neural Architecture Search, Qualcomm Inc.

HONORS

- Gold Reviewer (top 25%), ICML 2026 Jul 2026
- Google Conference Scholarship, NeurIPS 2024 draft-refinement paper Dec 2024
- Google Cloud Platform Credit Award, Gemma Academic Program Mar 2024
- 4th Place, NeurIPS Weather4Cast Competition; Project Lead Dec 2022
- Qualcomm AI Fellowship Oct 2022
- Best Poster Awards, KAIST AI Workshop (FINE; knowledge distillation) Jan 2022
- 8th Place, NeurIPS Black-Box Optimization Challenge; Project Lead Dec 2020
- 2nd & 3rd Places, NeurIPS MicroNet Challenge, CIFAR-100; Project Lead Dec 2019

LEADERSHIP

- Doctoral Student Representative, KAIST AI** Mar 2021 - Feb 2022
Built student organizations and served as liaison between faculty and students.
- Lab Master, OSI Lab, KAIST** Mar 2020 - Feb 2021
Organized seminars, lab policy, and funding processes; liaison between advisor and students.
- Vice Captain, ELKA (KAIST cheerleading group)** Oct 2014 - Nov 2015
KAIST festival planning and promotion; 1st prize at the 2015 Gwangju Summer Universiade cheering festival.

INVITED TALKS

• Naver AI Lab Seminar – LLM Inference-Time Strategies and Beyond	<i>Jan 2025</i>
• Microsoft Research Seminar – Instructive Decoding and Beyond	<i>Jan 2025</i>
• Squeezebits Efficient AI Seminar – Towards Fast LLM Inference	<i>Jun 2024</i>
• KCC 2024 – Semi-Supervised Federated Object Detection	<i>Jun 2024</i>
• HyperConnect Seminar – Personalized Federated Learning under Label Noise	<i>Jan 2023</i>
• KSC 2022 – FINE Samples for Learning with Noisy Labels	<i>Dec 2022</i>
• Naver Clova AI Seminar – Orthogonal Feature Regularization	<i>Jun 2019</i>

SERVICE & SKILLS

- **Reviewer:** NeurIPS 2022-2024; ICML 2022-2024, 2026; ICLR 2024-2025; AutoML-Conf 2022; ECCV 2022; KDD 2023; ARR 2024. Journal: Pattern Recognition.
- **Skills:** Python, PyTorch, LaTeX, vLLM/SGLang. LLM post-training (SFT, DPO, GRPO), inference optimization, retrieval, and agent tool orchestration.
- **AI in art:** Surplus Human, with Vania Oh and Junbeom Shin. AI with Weird Wonderland, CELINE PARK GALLERY, March 2022; organized by Next Interface Lab.

TEACHING

• AI Lecturer, ELICE – Mathematics in ML; lectures for LG employees	<i>Jan-Feb 2022</i>
• AI Mentor, Curinc – Introduction to Deep Learning; 70-hour course	<i>Jun-Jul 2021</i>
• TA, LG – Computer Vision and Deep Learning	<i>Oct-Nov 2020</i>
• Advisory Committee, National Science Museum – Deep Learning	<i>Mar-Oct 2020</i>
• TA, KAIST Data Science – Introduction to Deep Learning	<i>Mar-Jul 2019</i>
• TA, LG – Computer Vision and Deep Learning	<i>Jan 2019</i>
• Lecturer, Samsung – Python Basics	<i>Jan 2019</i>
• Lecturer, KAIST – Regular Leadership Course	<i>Mar-Dec 2017</i>

FUNDING

• KAIST Graduate School of AI doctoral funding: US\$20,000 per year during Ph.D. studies.	
• Korea Student Aid Foundation graduate scholarship: US\$16,000 per year	<i>Mar 2018 - Feb 2020</i>
• KAIST full academic scholarship: US\$3,000 per year	<i>Mar 2013 - Aug 2017</i>